Experiment 5
Naive Bayes Classification

Aim:

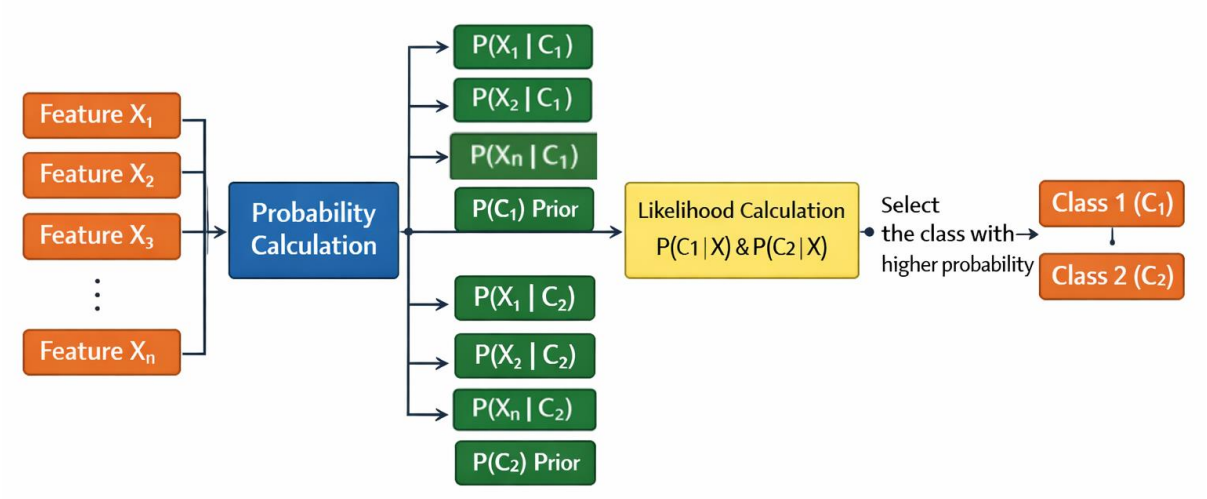
To implement the Naïve Bayes algorithm for predicting class labels and to evaluate its performance

Theory:

Naive Bayes is a supervised probabilistic learning algorithm widely used for text classification tasks such as spam and ham message detection. The model is based on Bayes' theorem, which estimates the posterior probability of a class given observed featur. A key assumption of Naive Bayes is that features are conditionally independent given the class label. While this assumption does not fully hold for natural language, the classifier remains effective due to the high dimensionality and sparse distribution of textual features.
Bayes' theorem is defined as:

$$P(C \mid X) = \frac{P(X \mid C)\, P(C)}{P(X)}$$

where $P(C \mid X)$represents the probability of a message belonging to class C(spam or ham) given the feature set X, $P(X \mid C)$is the likelihood of observing the features under class C, P(X) is the marginal likelihood of the features, and $P(C)$denotes the prior probability of the class. The feature-wise conditional probabilities and class priors are computed to select the class with the highest posterior probability is shown in the figure below:



In text classification, messages are modelled using the bag-of-words representation, where word order is ignored and word occurrences form the basis of feature extraction. To improve feature quality and reduce redundancy, Natural Language Processing (NLP) techniques are applied. These include text normalization, removal of stop words, and reduction of words to their base forms using stemming and lemmatization. Part-of-Speech (POS) tagging supports lemmatization by enabling accurate identification of grammatical roles such as nouns and verbs, thereby preserving semantic correctness while reducing dimensionality.

Textual data is transformed into numerical vectors using Term Frequency–Inverse Document Frequency (TF-IDF) weighting. TF-IDF enhances the importance of words that are frequent within a document but infrequent across the entire corpus, thus improving the discriminative ability of the feature space.
For classification, the Multinomial Naive Bayes algorithm is employed, as it is well suited for word-based features and TF-IDF representations. Model effectiveness is assessed using standard metrics such as accuracy, precision, recall, confusion matrix, and classification report.

Merits of Naïve Bayes:

- Naive Bayes is suitable when the number of features is very large, such as in text classification problems with thousands of words.
- Naive Bayes is suitable when fast training and prediction are required because of its low computational complexity.
- Even when only a small amount of training data is available, Naive Bayes can still give acceptable results.
- When the dataset contains unnecessary or extra features, Naive Bayes can handle them without much loss in performance.

Demerits of Naïve Bayes:

- When features are highly correlated or dependent, Naive Bayes should not be used because the independence assumption is violated.
- When the problem is complex and non-linear, Naive Bayes is not suitable since it cannot model complex patterns.

Overall, the Naive Bayes classifier, combined with NLP-based feature normalization and TF-IDF representation, provides an efficient and robust solution for spam and ham message classification.

Code and Results Section:

Block 1: Importing Required Libraries

Input

```
import pandas as pd

import numpy as np

import re

import matplotlib.pyplot as plt

import nltk

from nltk.stemimport PorterStemmer

from sklearn.feature_extraction.textimport ENGLISH_STOP_WORDS

from sklearn.model_selectionimport train_test_split

from sklearn.feature_extraction.text import TfidfVectorizer
```

```
from sklearn.naive_bayes import MultinomialNB

from sklearn.metrics import accuracy_score, classification_report, confusion_matrix, precision_recall_curve

print("Libraries Imported")
```

Output

Libraries Imported

Block 2: Reading the Dataset

Input

```
data = pd.read_csv('/content/Spam_Detection.csv', encoding='latin1', header=0, usecols=[0, 1], engine="python")

print("Dataset reading complete")
```

Output

Dataset reading complete

Block 3: Dataset Preview (First 5 rows)

Input

```
data.head()
```

Output

	Category	Message
0	spam	Get your garden ready for summer with a free selection of summer bulbs and seeds worth £33.50, available only with The Scotsman this Saturday. To stop messages, visit notxt.co.uk.
1	spam	This is your chance to be on a reality fantasy show. Call now at 08707509020. The cost is 20p per minute. NTT Ltd, PO Box 1327, Croydon CR9 5WB. 0870 is a national-rate call.
2	ham	Love is not a decision; it is a feeling. If we could choose whom to love, life would be much simpler, but it would also be less magical.
3	ham	Good friends care for each other, close friends understand each other, and true friends remain forever—beyond words and beyond time. Good night.
4	spam	Get your garden ready for summer with a free selection of summer bulbs and seeds worth £33.50, available only with The Scotsman this Saturday. To stop messages, visit notxt.co.uk.

Block 4: Dataset Preview (Last 5 rows)

Input

data.tail()

Output

	Category	Message
1737	spam	Do you want 750 anytime any network mins 150 text and a NEW VIDEO phone for only five pounds per week call 08002888812 or reply for delivery tomorrow
1738	ham	Thanx...
1739	spam	88800 and 89034 are premium phone services call 08718711108
1740	spam	Today's Offer! Claim ur £150 worth of discount vouchers! Text YES to 85023 now! SavaMob, member offers mobile! T Cs 08717898035. £3.00 Sub. 16 . Unsub reply X
1741	ham	am up to my eyes in philosophy

Block 5: Dataset Information

Input

data.info()

Output

Range Index: 1742 entries, 0 to 1741

Data columns (total 2 columns):

	Column	Non-Null Count	Dtype
0	Category	1742 non-null	object
1	Message	1742 non-null	object

dtypes: object(2)

memory usage: 27.3+ KB

Block 6: Missing Values

Input

data.isnull().sum()

Output

Category 0

Message 0

dtype: int64

Block 7: Size of the dataset

Input

data.shape

Output

(1742,2)

Block 8: Value count of each category

Input

data[‘Category'].value_counts()

Output

Category	Count
Ham	1003
Spam	739

Block 9: Visualizes the distribution of spam and ham messages in the dataset

Input

category_counts = data['Category’].value_counts()

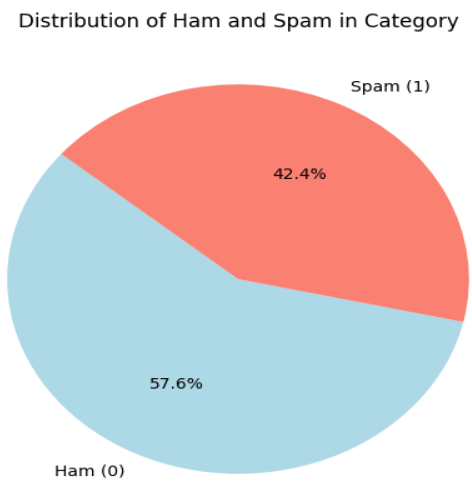
plt.figure(figsize=(6, 6))

plt.pie(category_counts, labels=['Ham (0)', 'Spam (1)'], autopct='%1.1f%%', startangle=140, colors=['lightblue', 'salmon'])

plt.title('Distribution of Ham and Spam in Category’)

plt.show()

Output



Block 10: Text preprocessing

Input

```
stemmer = PorterStemmer()
stop_words = set(ENGLISH_STOP_WORDS)
print("Initialized Porter Stemmer and English stopword list.")
```

Output

Initialized Porter Stemmer and English stopword list.

Block 11: Text cleaning and Normalization

Input

```
def preprocess_text(s):
    if pd.isnull(s):
        return ""
    s = s.lower()
    s = re.sub(r'http\S+|www\.\S+', ' ', s)
    s = re.sub(r'[^a-z0-9\s]', ' ', s)
    s = re.sub(r'\s+', ' ', s).strip()
    return s
print(“Text Cleaned”)
```

Output

Text Cleaned

Block 12: Splits the dataset

Input

```
data['label'] = (data['Category'].str.lower().str.strip() == 'spam').astype(int)

X_train, X_test, y_train, y_test = train_test_split( data['Message'], data['label'],

    test_size=0.25, random_state=42, stratify=data['label’] )

print("Labels encoded and data split into training and testing sets.")
```

Output

Labels encoded and data split into training and testing sets.

Block 13: Converts text messages into TF-IDF feature vectors

Input

```
vectorizer = TfidfVectorizer(ngram_range=(1,3), min_df=2, max_features=50000)

X_train_vec = vectorizer.fit_transform(X_train)

X_test_vec = vectorizer.transform(X_test)

print("Text data vectorized using TF-IDF.")
```

Output

Text data vectorized using TF-IDF.

Block 14: Multinomial Naive Bayes classifier model trains

Input

```
model=MultinomialNB()

model.fit(X_train_vec,y_train)
```

Output

```
MultinomialNB
MultinomialNB()
```

Block 15: Model Evaluation

Input

```
y_pred = model.predict(X_test_vec)
y_prob = model.predict_proba(X_test_vec)[:, 1]
print("Accuracy:", accuracy_score(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred))
```

Output

```
Accuracy: 0.9288990825688074
Classification Report:
```

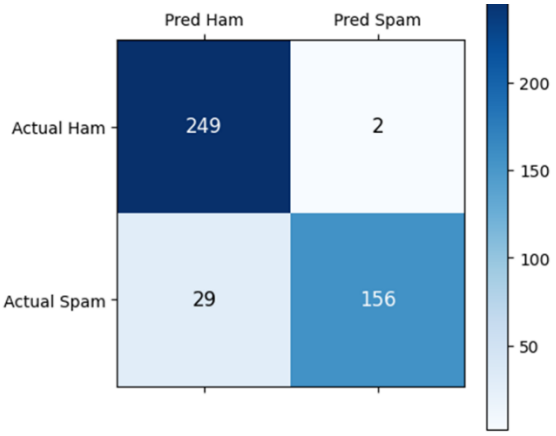
	precision	recall	f1-score	support
0(ham)	0.90	0.99	0.94	251
1(spam)	0.99	0.84	0.91	185
accuracy			0.93	436
macro avg	0.94	0.92	0.93	436
weighted avg	0.93	0.93	0.93	436

Block 16: Displays confusion matrix

Input

```
cm = confusion_matrix(y_test, y_pred)
plt.matshow(cm, cmap='Blues')
for (i, j), val in np.ndenumerate(cm):
    plt.text(j, i, val, ha='center', va='center',
             color='white' if val > cm.max()/2 else 'black', fontsize=12)
plt.xticks([0,1], ['Pred Ham','Pred Spam'])
plt.yticks([0,1], ['Actual Ham','Actual Spam'])
plt.title("Confusion Matrix", fontsize=14, fontweight='bold')
plt.colorbar()
plt.show()
```

Output

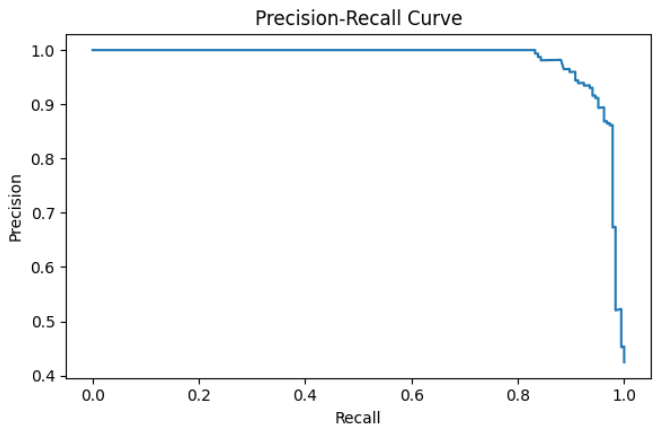


Block 17: Displays the Precision-Recall curve for the model

Input

```
precision, recall, _ = precision_recall_curve(y_test, y_prob)
plt.figure(figsize=(6,4))
plt.plot(recall, precision)
plt.title('Precision-Recall Curve')
plt.xlabel('Recall')
plt.ylabel('Precision')
plt.tight_layout()
plt.show()
```

Output



Block 18: Highlights words using color intensity

Input

```
from IPython.display import HTML, display

def color_text(words, contributions): max_val = max(abs(c) for c in contributions) or 1

 html = ""

   for w, c in zip(words, contributions): intensity = int(255 * (abs(c) / max_val))

      if c > 0: color = f"rgb(255,{255-intensity},{255-intensity})"

      else: color = f"rgb({255-intensity},255,{255-intensity})"

            html += f"<span style='background-color:{color}; padding:3px; margin:2px; border-radius:4px;'>{w}</span> "

return html

print("Text contribution visualization function defined.")
```

Output

Text contribution visualization function defined.

Block 19: Computes precision and recall of the model on test data

Input

```
def show_precision_recall(model, X_test_vec, y_test):

   preds = model.predict(X_test_vec)

   precision = precision_score(y_test, preds)

   recall = recall_score(y_test, preds)

print("Precision and recall calculation function ready.")
```

Output

Precision and recall calculation function ready.

Block 20: Function for 'n' random test messages

Input

```
def analyze_multiple(model, vectorizer, X_test, y_test, n=5):

   show_precision_recall(model, X_test_vec, y_test)

   for i in range(n):

      analyze_random_message(model, vectorizer, X_test, y_test)

print("Multiple message analysis function defined.")
```

Output

Multiple message analysis function defined.

Block 21: Evaluation on test messages

Input

analyze_multiple(model, vectorizer, X_test, y_test, n=5)

print(“ MODEL PERFORMANCE ON TEST DATA");

print("Precision:")

print("Recall:")

print(“ Choose one random test sample.”)

Output

MODEL PERFORMANCE ON TEST DATA

Precision: 0.9873

Recall: 0.8432

Choose one random test sample.

Category	Message
ham	It's getting me down just waiting around.
spam	Text and meet someone today. Reply with your name and age.
ham	I.ll give her once i have it. Plus she said grinule greet you whenever we speak
spam	Congratulations U can claim 2 VIP row A Tickets 2 C Blu in concert in November or Blu gift guaranteed Call 09061104276 to claim TS&Cs www.smsco.net cost£3.75max
ham	thinking about you.

Message Analysis

it's getting me down just waiting around.

Actual: ham
Predicted: ham
Spam Probability: 0.0408

Naive Bayes - Practical Implementation

Virtual Labs - Dayalbagh Educational Institute

Step 1: Importing Libraries

 # Importing necessary libraries

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import MultinomialNB
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
import seaborn as sns
import matplotlib.pyplot as plt
from IPython.display import HTML, display
print("Libraries Imported")
```

Output:

```
Libraries Imported
```

Step 2: Loading Dataset

 # Read the Spam SMS dataset

```
data = pd.read_csv('spam_detection.csv', encoding='latin1', header=0, usecols=[0,1],
    engine="python")
print("Dataset loaded successfully")
```

Output:

```
Dataset loaded successfully
```

Step 3: Data Analysis

 # Display first 5 rows

```
data.head()
```

Output:

```
Category  Message
0  spam      Get your garden re
1  spam      This is your chanc
2  ham       Love is not a deci
3  ham       Good friends care
4  spam      Get your garden re
```

 # Display Last 5 rows

```
data.tail()
```

Output:

```
Category  Message
1737  spam      Do you want 750 an
1738  ham       Thanx...
1739  spam      88800 and 89034 ar
1740  spam      Today's Offer! Cla
1741  ham       am up to my eyes i
```

Displays summary of dataset, including column names, data types, and non-null counts.

```
data.info()
```

Output:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1742 entries, 0 to 1741
Data columns (total 2 columns):
#   Column    Non-Null Count  Dtype
---  ------    --------------  -----
0   Category  1742 non-null   object
1   Message   1742 non-null   object
dtypes: object(2)
memory usage: 27.3+ KB
```

Checks the dataset for missing(NULL) values in each column

```
data.isnull().sum()
```

Output:

```
Category    0
Message     0
dtype: int64
```

Shows the size of the dataset

```
data.shape
```

Output:

```
(1742, 2)
```

Counts the frequency of each unique category

```
data['Category'].value_counts()
```

Output:

```
Category  Count
Ham       1003
Spam      739
dtype: int64
```

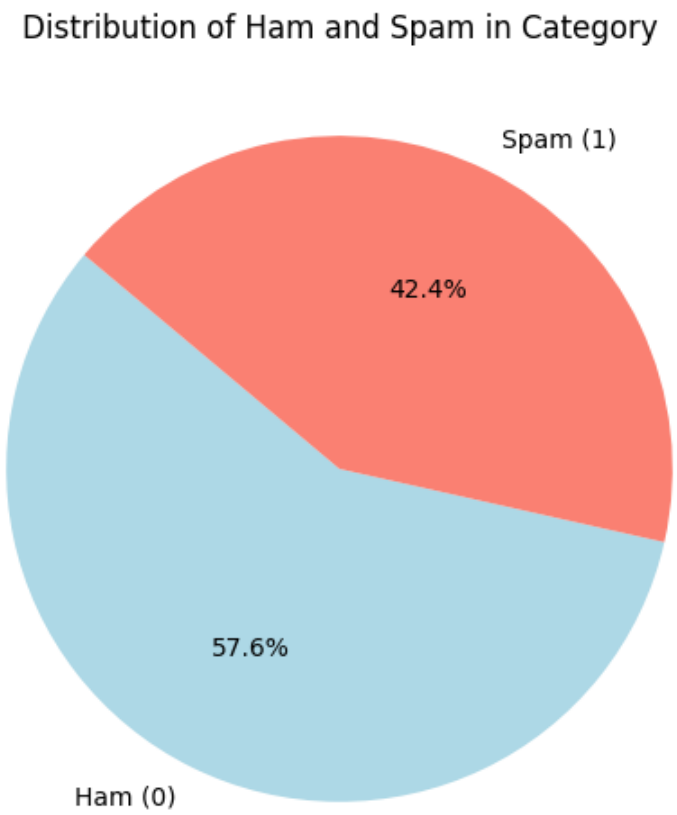
Step 4: Data Preprocessing

 # Visualizes the distribution of spam and ham messages in the dataset

```
category_counts = data['Category'].value_counts()
plt.figure(figsize=(6, 6))
plt.pie(category_counts, labels=['Ham (0)', 'Spam (1)'],
    autopct='%1.1f%%', startangle=140, colors=['lightblue', 'salmon'])
plt.title('Distribution of Ham and Spam in Category');
plt.show()
```

Output:

```
Distribution of Ham and Spam in Category
```



 # Sets up stemming and stop-word removal tools for text preprocessing.

```
stemmer = PorterStemmer()
stop_words = set(ENGLISH_STOP_WORDS)
print("Initialized Porter Stemmer and English stopword list.")
```

Output:

```
Initialized Porter Stemmer and English stopword list.
```

 # Performs basic text cleaning and normalisation

```
def preprocess_text(s):
    if pd.isnull(s): return ""
    s = s.lower(); s = re.sub(r'http\S+|www\.\S+', ' ', s); s = re.sub(r'[^a-z0-9\s]', '
    ', s)
    s = re.sub(r'\s+', ' ', s).strip(); return s
print("Text Cleaned")
```

Output:

```
Text Cleaned
```

Step 5: Model Training

 # Encodes spam labels and splits the data into stratified training and testing sets.

```
data['label'] = (data['Category'].str.lower().str.strip() == 'spam').astype(int)
X_train, X_test, y_train, y_test = train_test_split( data['Message'], data['label'],
    test_size=0.25, random_state=42, stratify=data['label'] )
print("Labels encoded and data split into training and testing sets.")
```

Output:

```
Labels encoded and data split into training and testing sets.
```

 # Converts text messages into TF-IDF feature vectors.

```
vectorizer = TfidfVectorizer(ngram_range=(1,3), min_df=2, max_features=50000)
X_train_vec = vectorizer.fit_transform(X_train)
X_test_vec = vectorizer.transform(X_test)
print("Text data vectorized using TF-IDF.")
```

Output:

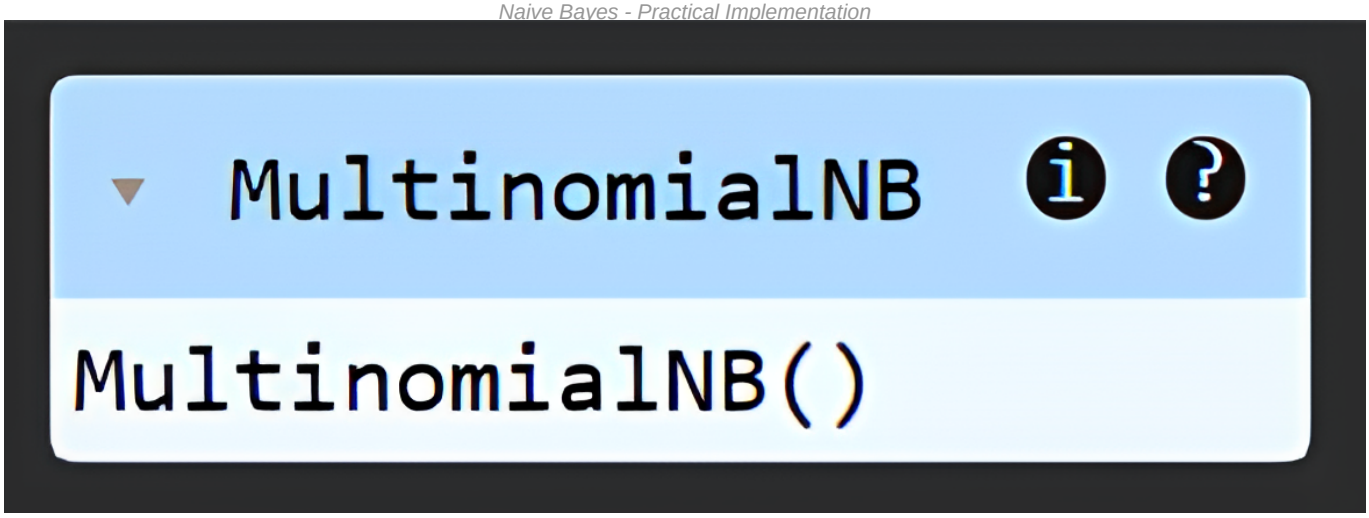
```
Text data vectorized using TF-IDF.
```

 # Trains a Multinomial Naive Bayes classifier on the TF-IDF features.

```
model=MultinomialNB()
model.fit(X_train_vec,y_train)
print("Model training completed.")
```

Output:

```
Model training completed.
```



Step 6: Model Evaluation

 # Predicts on test data and evaluates model

```
y_pred = model.predict(X_test_vec)
y_prob = model.predict_proba(X_test_vec)[:, 1]
print("Accuracy:", accuracy_score(y_test, y_pred))
print("\\nClassification Report:\\n", classification_report(y_test, y_pred))
```

Output:

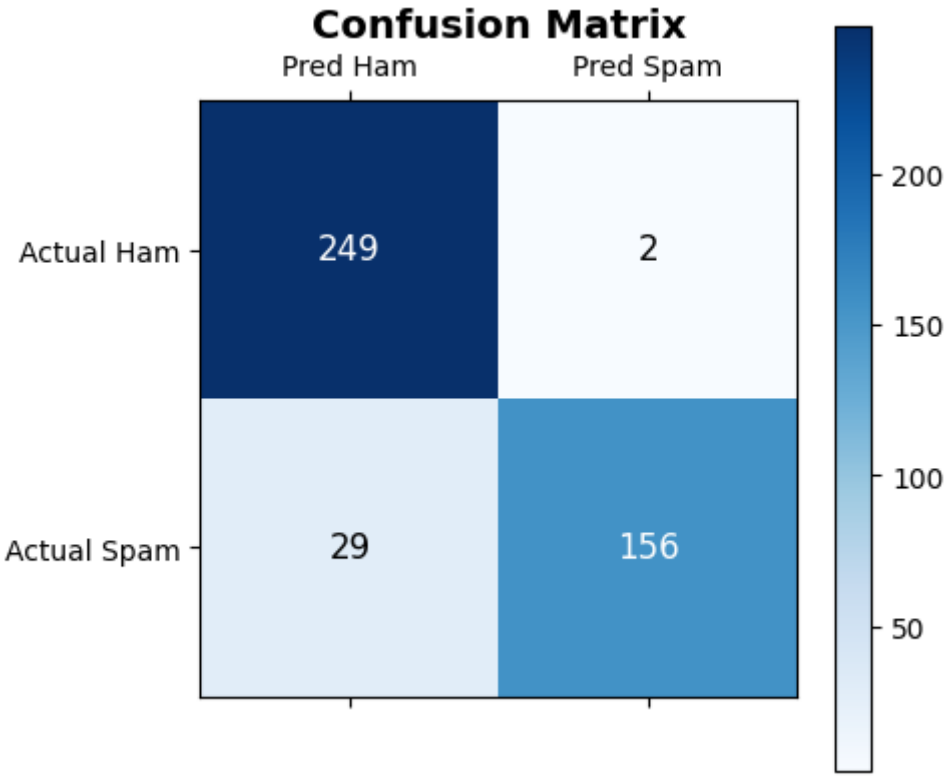
```
Accuracy: 0.9288990825688074
Classification Report:
precision    recall  f1-score   support
0(ham)       0.90      0.99      0.94       251
1(spam)       0.99      0.84      0.91       185
accuracy                           0.93       436
macro avg       0.94      0.92      0.93       436
weighted avg       0.93      0.93      0.93       436
```

 # Plots the confusion matrix with values to visualize correct and incorrect predictions

```
cm = confusion_matrix(y_test, y_pred); plt.matshow(cm, cmap='Blues')
for (i, j), val in np.ndenumerate(cm):
    plt.text(j, i, val, ha='center', va='center', color='white' if val > cm.max()/2 else
    'black', fontsize=12)
plt.xticks([0,1], ['Pred Ham','Pred Spam'])
plt.yticks([0,1], ['Actual Ham','Actual Spam']); plt.colorbar();
plt.show(); print("Confusion matrix displayed.")
```

Output:

```
Confusion matrix displayed.
```

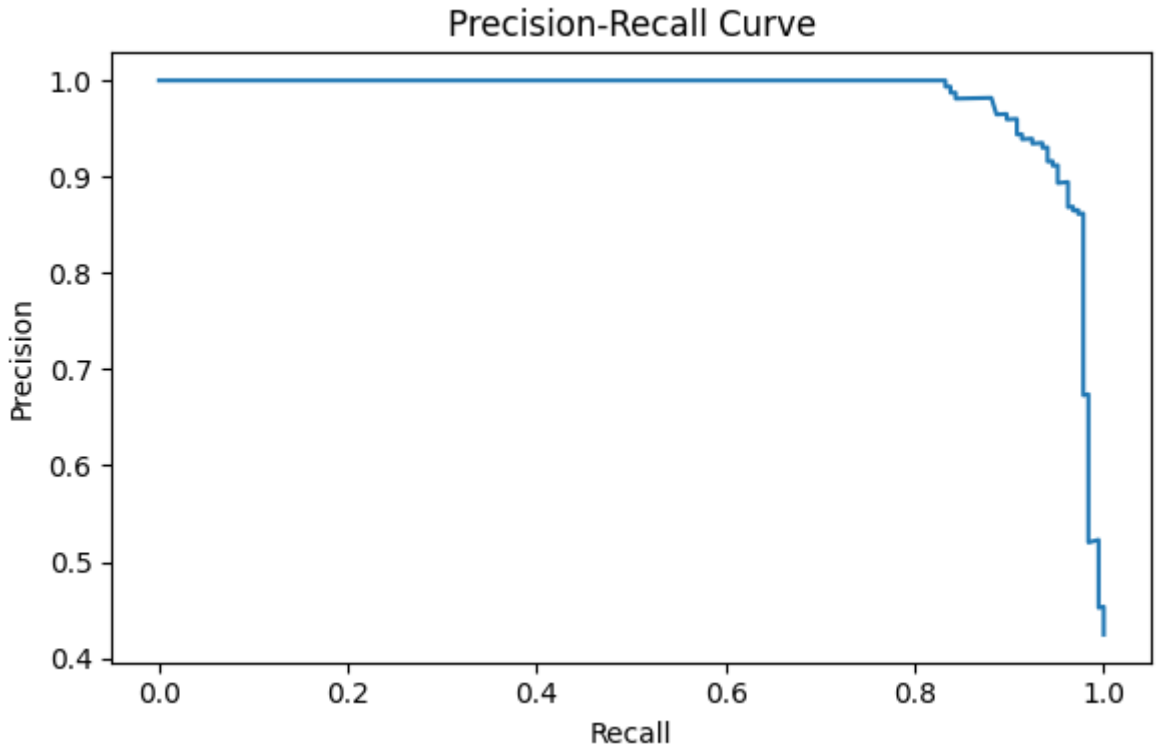


 # Plots the Precision-Recall curve for the model

```
precision, recall, _ = precision_recall_curve(y_test, y_prob)
plt.figure(figsize=(6,4)); plt.plot(recall, precision); plt.title('Precision-Recall
    Curve'); plt.xlabel('Recall');plt.ylabel('Precision'); plt.tight_layout(); plt.show()
print("Precision-Recall curve displayed.")
```

Output:

```
Precision-Recall curve displayed.
```



 # Highlights words based on their contribution to spam/ham prediction using color intensity

```
def color_text(words, contributions):
    max_val = max(abs(c) for c in contributions) or 1; html = ""
    for w, c in zip(words, contributions):
        intensity = int(255 * (abs(c) / max_val))
        if c > 0: color = f"rgb(255,{255-intensity},{255-intensity})"
        else: color = f"rgb({255-intensity},255,{255-intensity})"
        html += f"{w} "
    return html
print("Text contribution visualization function defined.")
```

Output:

```
Text contribution visualization function defined.
```

 # Computes and displays precision and recall of the model on test data

```
def show_precision_recall(model, X_test_vec, y_test):
    preds = model.predict(X_test_vec)
    precision = precision_score(y_test, preds) ;recall = recall_score(y_test, preds)
print("Precision and recall calculation function ready.")
```

Output:

```
Precision and recall calculation function ready.
```

 # Performs precision-recall evaluation and analyzes 'n' random test messages

```
def analyze_multiple(model, vectorizer, X_test, y_test, n=5):
    show_precision_recall(model, X_test_vec, y_test)
    for i in range(n):
        analyze_random_message(model, vectorizer, X_test, y_test)
print("Multiple message analysis function defined.")
```

Output:

```
Multiple message analysis function defined.
```

Runs evaluation and displays analysis of 5 random test messages

```
analyze_multiple(model, vectorizer, X_test, y_test, n=5)
print(" MODEL PERFORMANCE ON TEST DATA")
print("Precision:", round(precision, 4))
print("Recall:", round(recall, 4))
```

Output:

```
MODEL PERFORMANCE ON TEST DATA
Precision: 0.9873
Recall   : 0.8432
Choose one random test sample.
Category  Message
ham       It's getting me do
spam      Text and meet some
ham       I.ll give her once
spam      Congratulations U
ham       thinking about you
Message Analysis
Actual:
Predicted:
Spam Probability:
```